

ShuttleSense

ML for Robotics Semester Project

Presented By

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AGENDA

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Introduction

Problems

- Collecting shuttlecocks manually → more time/work
- Inconsistent Collection
- Labor Dependency

Solution

- Use of Deep Learning enables accurate object detection.
- Integration of Computer Vision with robotics.
- System uses camera input + trained model for detection / collecting

Methodology

- Used YOLOv8 for real-time object detection.
- Initialized model with pretrained weights from COCO Dataset.
- Applied transfer learning to adapt model for shuttlecock detection.
- Evaluated performance using: Precision & Recall, mAP (mean Average Precision).
- Integrated trained model with simulation for testing

Dataset

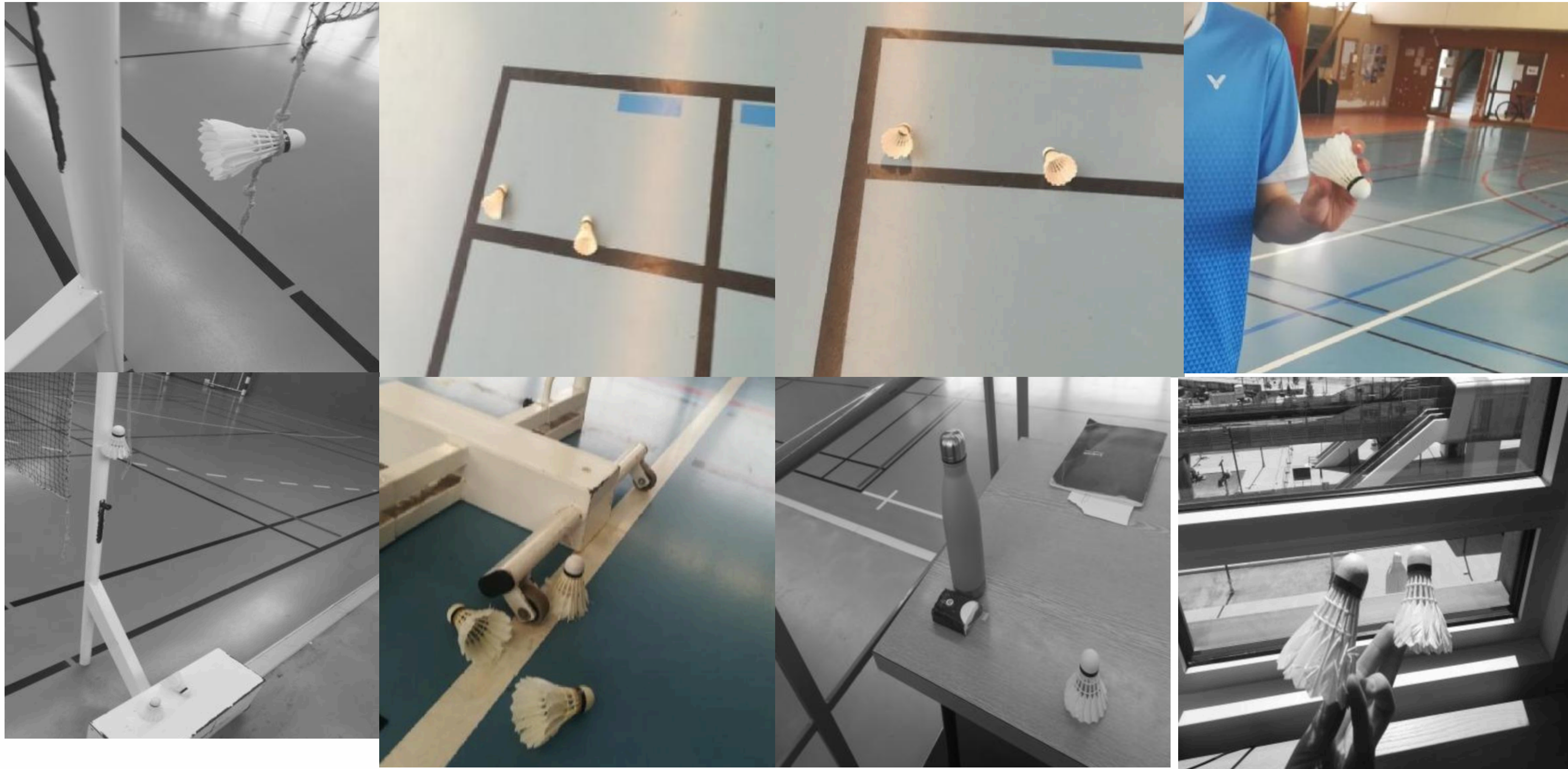
- YOLOv8 was pretrained on COCO dataset.
- Model was initially trained on COCO dataset.
- Helped the model learn strong general object detection features.
- Used Shuttle Badminton Photos Dataset from Kaggle for fine-tuning.

15,000 images

15,000 bounding box labels

Dataset (2)

Shuttlecock dataset from Kaggle used for fine-tuning



Data Pre-Processing

- Dataset was already structured into:
 - Training set
 - Validation set
 - Test set
- Labels were provided in YOLO format (bounding box coordinates already available).
- Used YOLOv8 built-in preprocessing:
 - Image resizing
 - Normalization
 - Data augmentation (flip, scale, rotation, etc.)

Data Pre-Processing (2)

Additional Preprocessing

- Removed incorrect and noisy labels
- Eliminated duplicate and blurred images

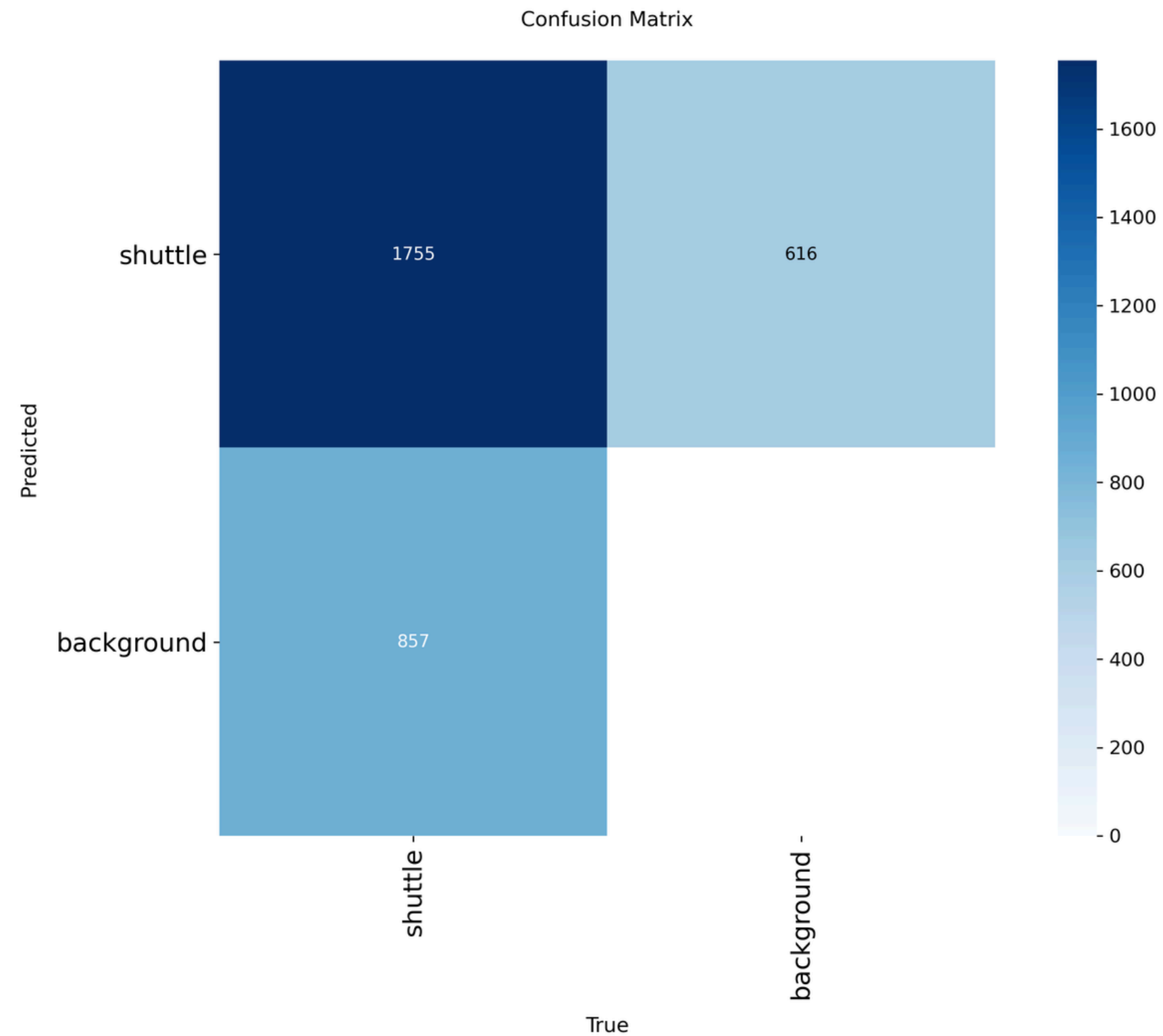
Core preprocessing was already handled by dataset structure and YOLO pipeline

Results & Analysis

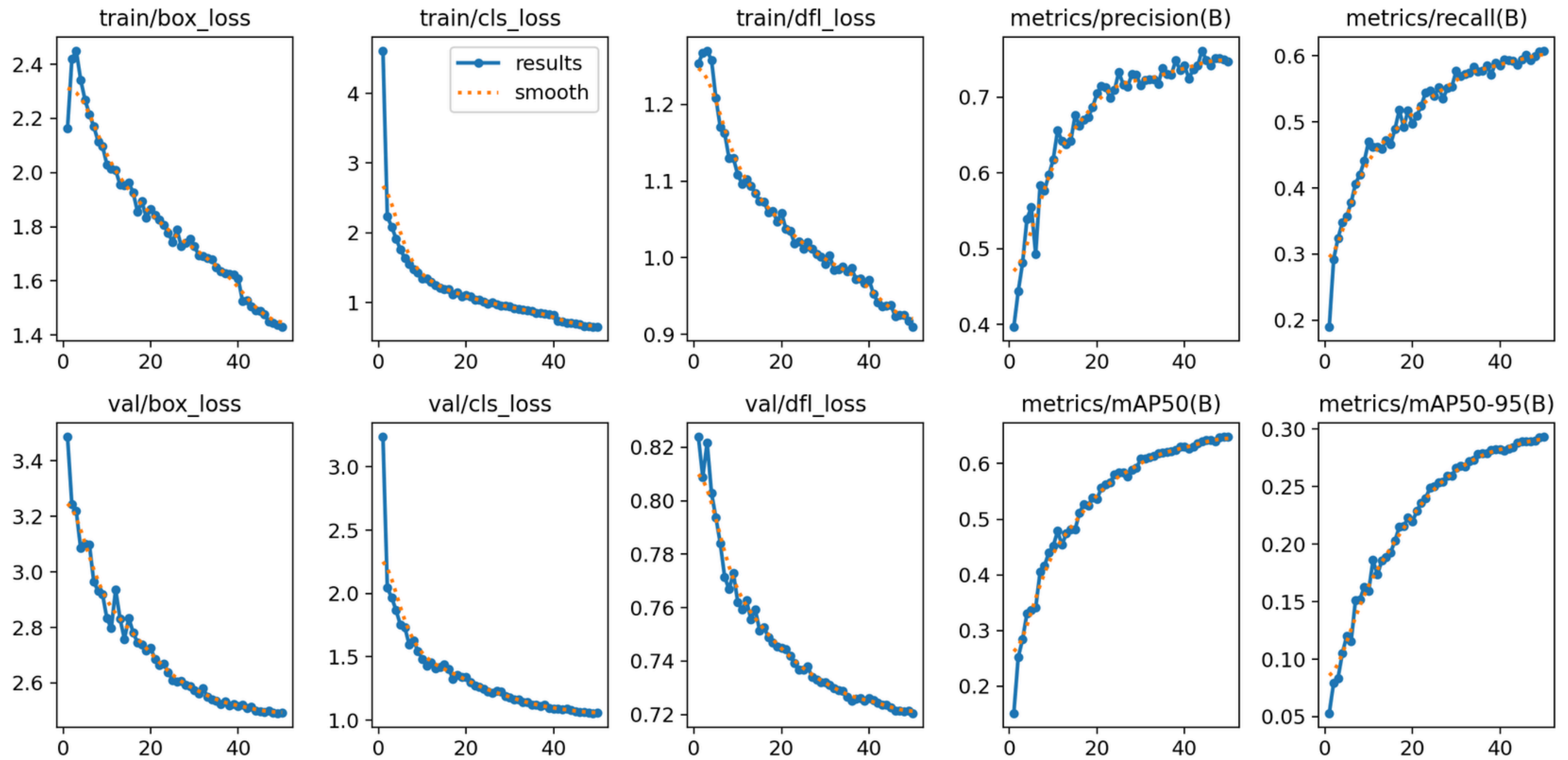
Epoch	Precision	Recall	mAP@50
1	39.7%	19.0%	15.1%
25	73.3%	53.9%	58.3%
50	74.6%	60.8%	64.8%

The model improved rapidly in the early epochs and then gradually plateaued toward the end, which is typical behavior.

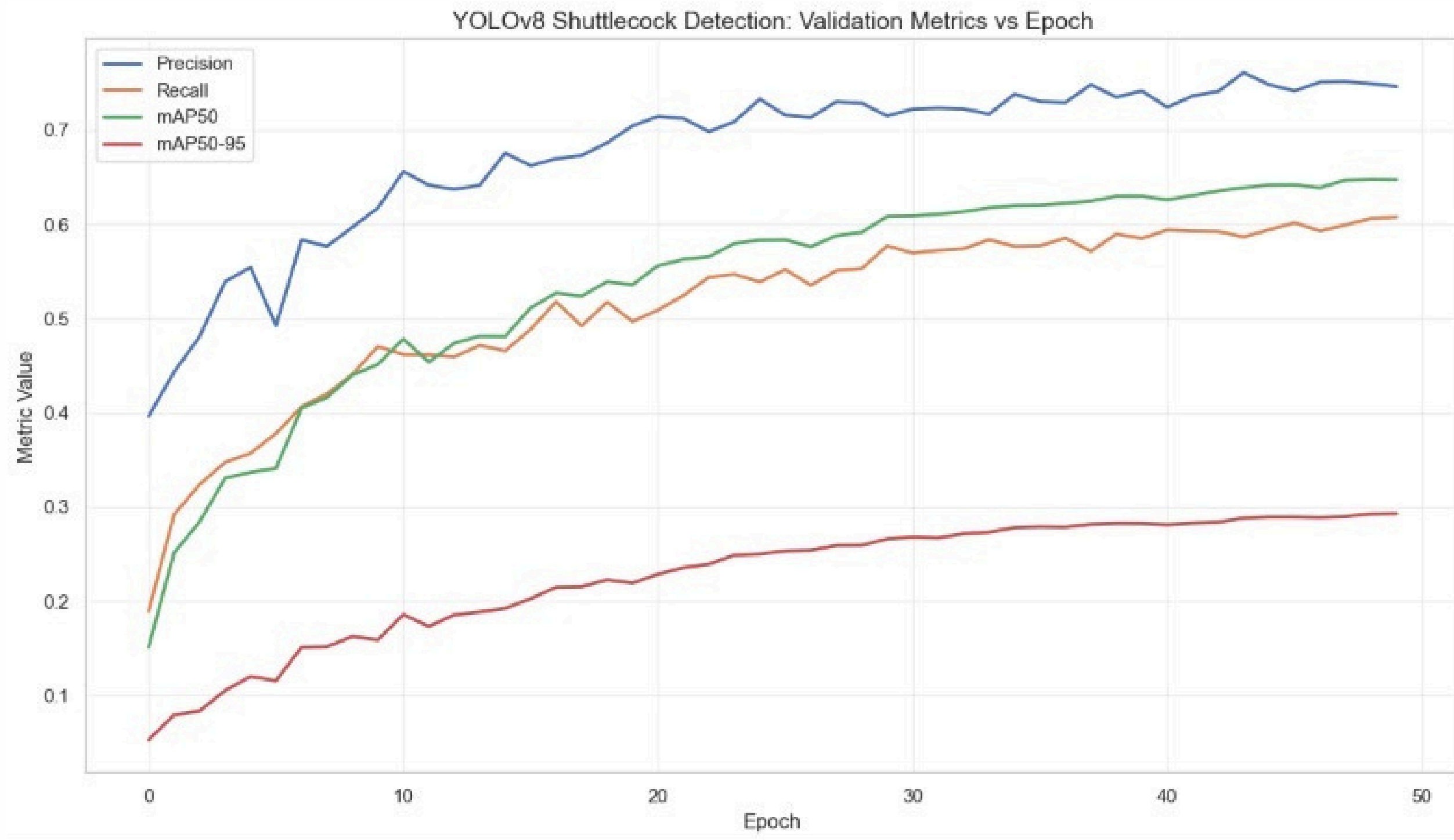
Results & Analysis



Results & Analysis



Results & Analysis



Results & Analysis

Original Image



Shuttlecock Detection using YOLOv8



References

- <https://www.kaggle.com/datasets/ayushsinha731/shuttle-badminton-photos>
- <https://www.kaggle.com/datasets/awsaf49/coco-2017-dataset>



ShuttleSense **Simulation**



Thank You